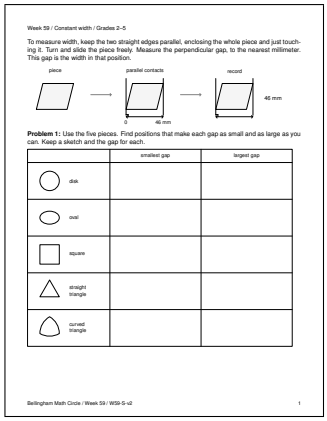
Week 59 / Constant width / Page menu

For adults. Band slices print the pages whose range includes the band, with page and problem numbers unchanged.
K–1: no pages · Grades 2–3: pp. 1–5 (5 of 6) · Grades 4–5: whole packet


Week 59 / Constant width / Grades 2–5

To measure width, keep the two straight edges parallel, enclosing the whole piece and just touching it. Turn and slide the piece freely. Measure the perpendicular gap, to the nearest millimeter. This gap is the width in that position.

piece · parallel contacts · record

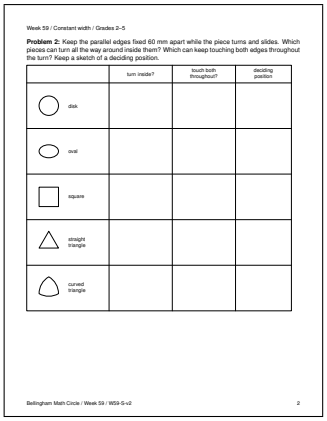
46 mm

Problem 1: Use the five pieces. Find positions that make each gap as small and as large as you can. Keep a sketch and the gap for each.

	smallest gap	largest gap
disk		
oval		
square		
straight triangle		
curved triangle		

Bellingham Math Circle / Week 59 / W59-S-v2 1

page 1 · 2–5 · P1

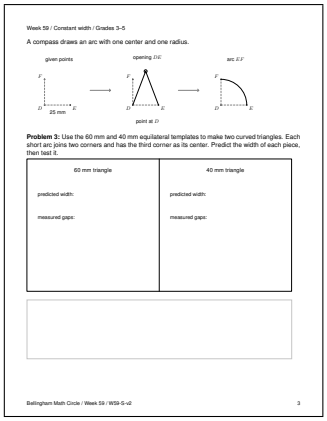

Week 59 / Constant width / Grades 2–5

Problem 2: Keep the parallel edges fixed 60 mm apart while the piece turns and slides. Which pieces can turn all the way around inside them? Which can keep touching both edges throughout the turn? Keep a sketch of a deciding position.

	turn inside?	touch both throughout?	deciding position
disk			
oval			
square			
straight triangle			
curved triangle			

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page 2 · 2–5 · P2


Week 59 / Constant width / Grades 3–5

A compass draws an arc with one center and one radius.

given points · opening DE · arc EF

25 mm

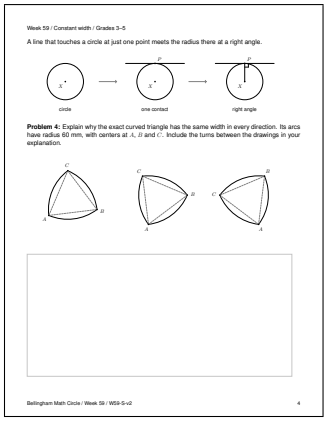
point at D

Problem 3: Use the 60 mm and 40 mm equilateral templates to make two curved triangles. Each short arc joins two corners and has the third corner as its center. Predict the width of each piece, then test it.

60 mm triangle	40 mm triangle
predicted width:	predicted width:
measured gaps:	measured gaps:

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page 3 · 3–5 · P3


Week 59 / Constant width / Grades 3–5

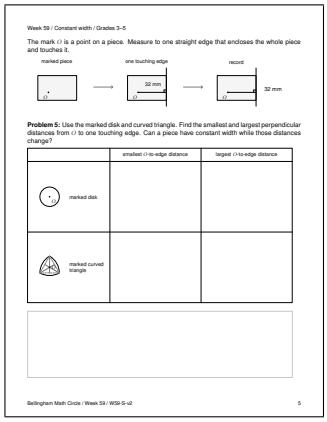
A line that touches a circle at just one point meets the radius there at a right angle.

circle · one contact · right angle

Problem 4: Explain why the exact curved triangle has the same width in every direction. Its arcs have radius 60 mm, with centers at A, B and C. Include the turns between the drawings in your explanation.

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page 4 · 3–5 · P4


Week 59 / Constant width / Grades 3–5

The mark O is a point on a piece. Measure to one straight edge that encloses the whole piece and touches it.

marked piece · one touching edge · record

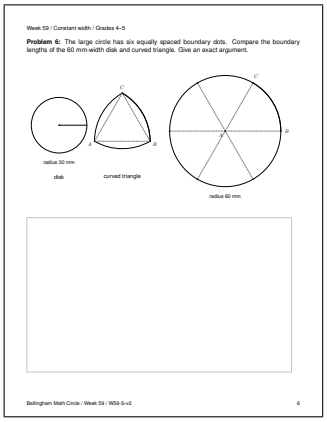
32 mm · 32 mm

Problem 5: Use the marked disk and curved triangle. Find the smallest and largest perpendicular distances from O to one touching edge. Can a piece have constant width while those distances change?

	smallest O-to-edge distance	largest O-to-edge distance
marked disk		
marked curved triangle		

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page 5 · 3–5 · P5


Week 59 / Constant width / Grades 4–5

Problem 6: The large circle has six equally spaced boundary dots. Compare the boundary lengths of the 60 mm-width disk and curved triangle. Give an exact argument.

radius 30 mm · disk · curved triangle · radius 60 mm

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page 6 · 4–5 · P6